

# Distinct dependent Mahler $U_1$ -numbers

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Conjecture 00000008326

**Result.** The universal algebraic-independence assertion is false: two distinct Liouville numbers can differ by one.

**Scope.** Both source versions assert that two  $U$ -numbers are universally algebraically independent. We refute this conjunct using two numbers in the standard Mahler class  $U_1$ . No conclusion about the other conjectured laws is needed. In Mahler's classification, the Liouville numbers are precisely the  $U_1$ -numbers; see Adamczewski–Bugeaud, Definition 1.1 and the following sentence, p. 884 [1]. This standard classification is used here, independently of the source's separate designation of Liouville numbers as  $U_\infty$ .

**The actual numbers.** A real number  $x$  is Liouville when, for every  $N \in \mathbb{N}$ , there are integers  $a, b$  with  $b > 1$  such that

$$0 < \left| x - \frac{a}{b} \right| < \frac{1}{b^N}.$$

Take the convergent factorial series

$$L = \sum_{n=0}^{\infty} \frac{1}{10^{n!}}, \quad M = L + 1.$$

The index starts at zero. This is the base-ten Liouville series with its extra initial  $1/10$  term, which does not affect the approximation property. For completeness, its truncation through  $n = k$  has denominator  $b = 10^{k!}$  and integral numerator. For  $k \geq \max(1, N)$ , its positive tail is bounded by

$$\sum_{n=k+1}^{\infty} 10^{-n!} \leq \frac{10}{9} 10^{-(k+1)!} < 10^{-Nk!} = b^{-N}.$$

Indeed the factorial exponents after  $(k+1)!$  increase by at least one, and  $((k+1) - N)k! \geq 1$ . Thus  $L$  is Liouville.

For every approximant  $a/b$  of  $L$ , the integer numerator  $a + b$  gives

$$\left| M - \frac{a+b}{b} \right| = \left| L - \frac{a}{b} \right|.$$

The strict positive error and the denominator are unchanged. Hence  $M$  is also Liouville. Both numbers are individually transcendental, and  $L \neq M$ .

**Algebraic dependence.** The rational polynomial

$$P(X, Y) = Y - X - 1 \in \mathbb{Q}[X, Y]$$

is nonzero (its value at  $(0, 0)$  is  $-1$ ), whereas  $P(L, M) = 0$ . The pair is therefore not algebraically independent over  $\mathbb{Q}$ . This supplies two distinct  $U_1$ -numbers, hence two  $U$ -numbers, contradicting the universal conjunct.

**Formal verification.** The Lean project uses Mathlib’s actual `liouvilleNumber` infinite sum, proves its summability, and applies the verified `liouville_liouvilleNumber` approximation theorem. It proves preservation of the full `Liouville` predicate under translation, individual transcendence, and distinctness. The relation is an actual `MvPolynomial (Fin 2) ℚ`; its nonzero value and vanishing evaluation refute Mathlib’s `AlgebraicIndependent` predicate. The concluding existential theorem uses the Liouville approximation characterization of  $U_1$ , not a formalization of all Mahler classes. No numerical approximations or custom axioms are used.

## References

- [1] B. Adamczewski and Y. Bugeaud, *Transcendence measures for continued fractions involving repetitive or symmetric patterns*, J. Eur. Math. Soc. **12** (2010), 883–914, p. 884. doi:10.4171/JEMS/218.